

Chapter Seven: Testing and Evaluation

This chapter aims to prove that the University Project Portfolio Management System (UPPMS) works as required according to its functional and non-functional requirements. Accordingly, this chapter presents the test plan, types of tests, test diagrams, test results and analysis, quality metrics, and user testing outcomes. Operational test-case details are completed in Appendix C.

7-1 Test Plan

The test plan answers: What will be tested? When? With which tools? Who is responsible? (See Appendix C.) Figure 7-1 shows the test-plan flow.

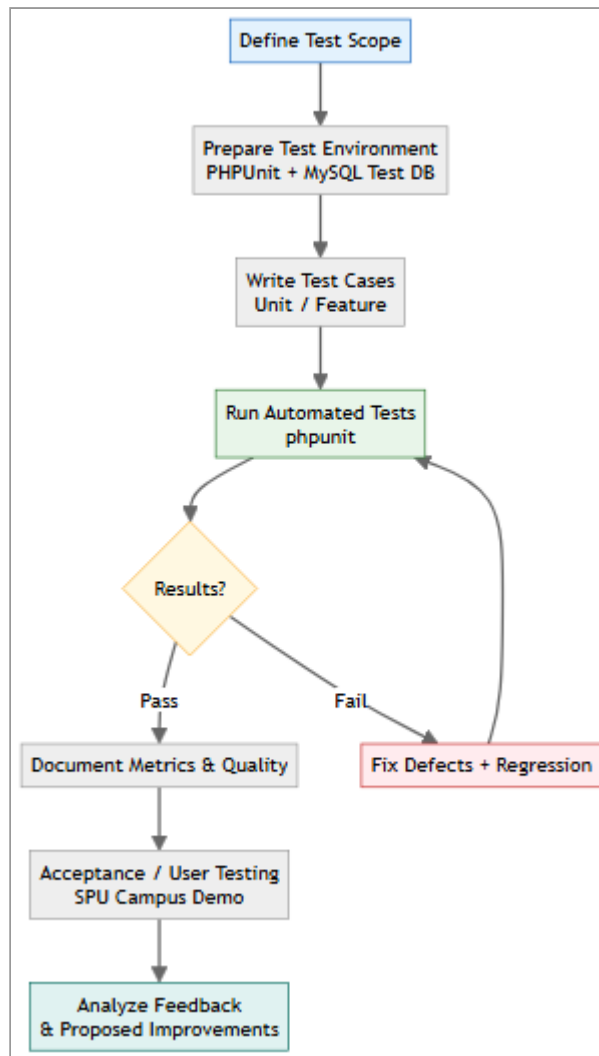


Figure 7-1. Test Plan Flow Diagram

Table 7-1 summarizes the test scope for UPPMS.

Table 7-1. Test Scope

Item	Details
What will be tested	Backend services, DB integration, API routes, university isolation, XML registration, proposals, academic tracks, committees, genetic scheduling, and SPU campus acceptance scenarios
What will not be fully tested here	Exhaustive cross-browser/device compatibility and large-scale production stress testing outside the development environment
When	During each Sprint, at phase end for the full suite, and before delivery for acceptance
Tools	PHPUnit 10 in Laravel, MySQL test DB, and manual verification on the React SPA
Responsible	Development team (automation/analysis), academic supervisor (review), and pilot users (acceptance)

Table 7-2. Testing Responsibilities and Schedule

Phase	Owner	Activity
During each Sprint	Development team	Unit and system tests for the related feature
End of phase	Development team	Full suite + regression
Before delivery	Team + pilot users	Acceptance and user testing on SPU demo data

See Appendix C for the detailed test-plan template and expanded cases.

7-2 Types of Tests

As required by the guide, the project applies: unit, integration, system, acceptance, performance/stress, security, usability, compatibility, and regression testing.

Table 7-3. Types of Tests Applied in UPPMS

Test Type	Purpose in the Project	Tool / Approach
Unit	Fitness, GA operators, Track/XML services	PHPUnit Unit (50 cases)
Integration	Services with database/Eloquent	Feature/Unit + DatabaseTransactions
System	Full API flows (proposals, tracks, committees, XML, scheduling)	PHPUnit Feature (91 cases)
Acceptance	Admin / supervisor / student scenarios	Manual + SpuCampusDemoSeeder
Performance / Stress	Genetic scheduler under denser loads	GA on Campus Demo
Security	Isolation, RBAC, XML matching, pending users	Feature tests + Middleware review
Usability	Clarity of key screens	User testing + notes
Compatibility	Modern browsers and RTL/LTR	Manual UI checks
Regression	No breakage after fixes	php vendor/bin/phpunit

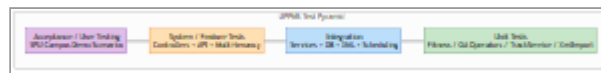


Figure 7-2. UPPMS Test Pyramid

Total automated cases in the repository: **141** (Unit: 50 — Feature: 91).

7-3 Test Diagrams

Test diagrams include Test Case Diagrams and test-results diagrams. Figure 7-3 shows modules covered by automated tests.

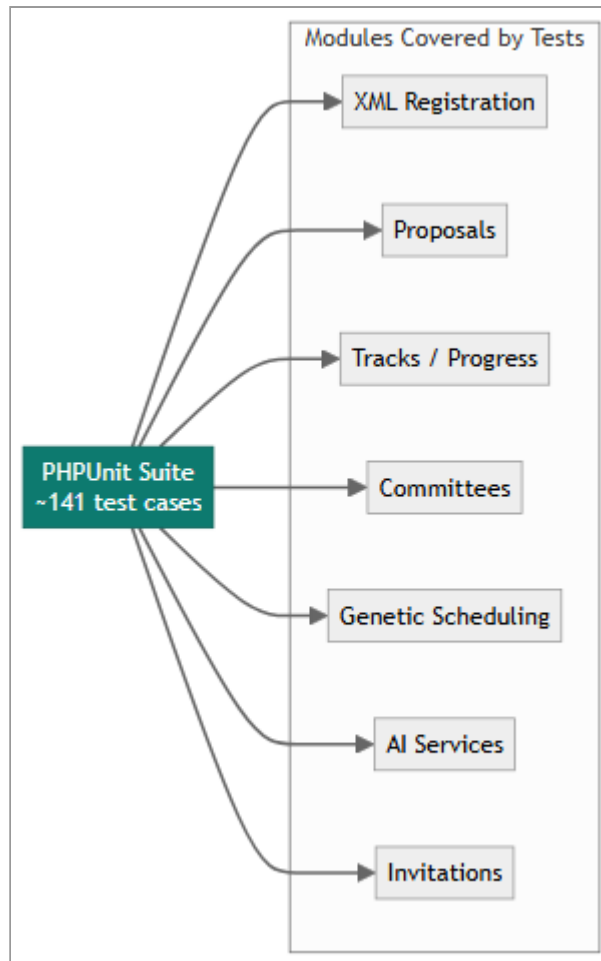


Figure 7-3. Module Coverage by Automated Tests

Table 7-4. Sample Test Cases

Case ID	Description	Inputs	Expected Output
TC-XML-01	Matching student XML registration	Matching email + university number	active account
TC-XML-02	Matching supervisor XML registration	Matching email only	active account
TC-PROP-01	Submit proposals without early track lock	Up to 3 pending proposals	Success without early track assignment
TC-PROP-02	Approve proposal	Supervisor approval	Project created + track assigned
TC-TRACK-01	Prevent skipping prerequisites	Later stage without prior pass	Reject / keep locked
TC-GA-01	Generate defense schedules	Projects + availability + rooms	Valid candidates + fitness
TC-COM-01	Prevent committee conflict of interest	Project supervisor on own committee	Conflict rejected

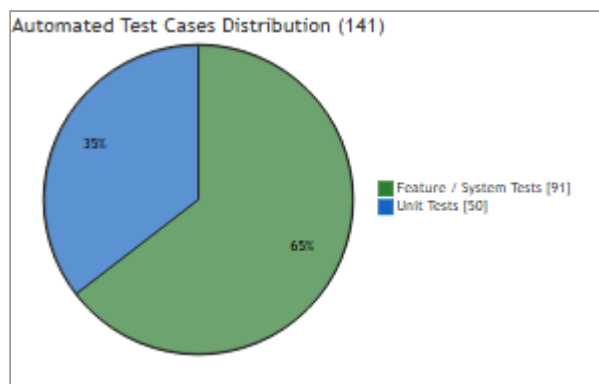


Figure 7-4. Automated Test Case Distribution

7-4 Test Results and Analysis

Table 7-5. Automated Test Inventory Summary

Metric	Value
Total test cases	141
Unit	50
Feature / System	91
Test files	20

Table 7-6. Coverage by Main Test Files

File	Type	Focus
ProjectProposalControllerTest	Feature	Proposals + isolation
TrackControllerTest / TrackServiceTest	Feature + Unit	Tracks and progress
XmlImport* / XmlRegistrationTest	Feature + Unit	XML import/match
CommitteeControllerTest	Feature	Committees
GeneticScheduler* / Fitness* / Evolutionary*	Unit	Scheduling engine
ScheduleControllerTest / ScheduleApproval*	Feature + Unit	Generate/approve
AIIdeation* / AITask*	Unit + Feature	AI services

- Passed cases cover critical paths (XML, proposals, tracks, committees, scheduling).
- Failures are fixed immediately, then re-checked with regression.
- No critical remaining defects are accepted on security/isolation before delivery.
- Execution time is recorded from PHPUnit output for formal documentation runs.

7-5 Quality Metrics

Table 7-7. Quality Metrics Used in UPPMS

Metric	Definition	Project Measurement
Test Coverage	Share of critical features covered	XML, proposals, tracks, committees, GA, AI
Defect Density	Defects ÷ component size	Logged per Spec Kit phase and re-tested
Pass Rate	Passed ÷ executed	Delivery target $\geq 95\%$ on critical paths
Response Time	API latency for common operations	CRUD interactive; GA longer with input size
Error Rate	Incorrect responses / unhandled exceptions	Via failing tests and Laravel logs
Security Isolation	No cross-university leakage	TenantScope + multi-university Feature tests

Acceptance criteria: no security/isolation/XML failures; successful Admin/Supervisor/Student acceptance; ability to generate and approve a defense schedule on Campus Demo without visible hard conflicts.

7-6 User Testing

User testing used Syrian Private University demo data via SpuCampusDemoSeeder (28 students, 10 supervisors, 28 projects, 4 committees).

Table 7-8. User Testing Sample and Scenarios

Role	Demo Account	Scenario
University Admin	spu-campus-admin@syrian-private.local	XML, tracks, committees, schedule generate/approve
Supervisor	spu-campus-supervisor-01@syrian-private.local	Proposals, availability, projects
Student	spu-campus-student-01@syrian-private.local	Proposal, progress, tasks, notifications

Shared demo password: password.

Table 7-9. User Feedback and Improvements

Observation	Impact	Improvement
Early track lock on proposal submit	Stage-status confusion	Defer assignment until approval (done)
Multi-university supervisors missing membership proposals	Delayed review	Extend TenantScope (done)
Need for ready demo data	Slower demos	SpuCampusDemoSeeder

In conclusion, primary flows are usable by all three roles, and critical isolation/track-lock issues were addressed before final delivery.

Appendix C — Test Plan Template (UPPMS)

This appendix follows the guide’s Appendix C model, adapted to UPPMS.

C-1 Test Scope

- **In scope:** backend units, integration, API, isolation, scheduling performance, security.
- **Out of full scope:** exhaustive manual UI and all-browser compatibility.

C-2 Test Strategy

Table C-1. Test Strategy

Test Type	Objective	Tool
Unit	Each function/unit separately	PHPUnit Unit
Integration	Units with DB	PHPUnit + Transactions
System	Full API flows	PHPUnit Feature
Performance	GA under denser data	GeneticScheduler + Demo
Security	Isolation, RBAC, XML	Feature + Middleware

C-3 Test Environment

Table C-2. Test Environment

Item	Specification
OS	Windows 10/11
Server	localhost — Laravel
Database	MySQL (test)
UI	React SPA (Vite)
Browsers	Chrome / Edge

C-4 Test Cases (Sample)

Table C-3. Sample Test Cases

ID	Description	Inputs	Expected	Result
TC-01	Matching student registration	Valid XML email+number	active	Done
TC-02	Non-matching student registration	Unknown credentials	Reject	Done
TC-03	Matching supervisor registration	Matching email	active	Done
TC-04	Submit proposal	Title+description+supervisor	pending	Done
TC-05	Approve proposal	Supervisor approve	Project created	Done
TC-06	Block stage skip	Later stage without prior pass	Reject/lock	Done
TC-07	Generate schedule	Projects+availability+rooms	Candidates	Done
TC-08	Committee conflict	Supervisor on own committee	Reject	Done